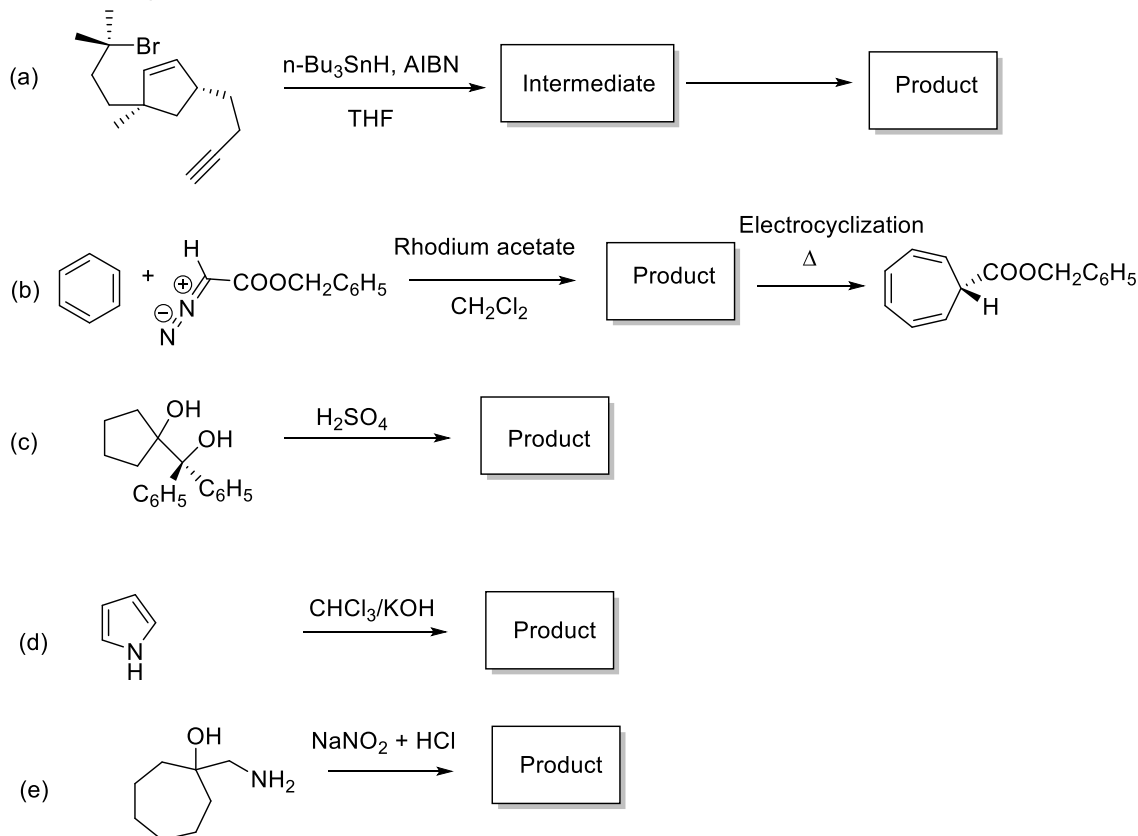


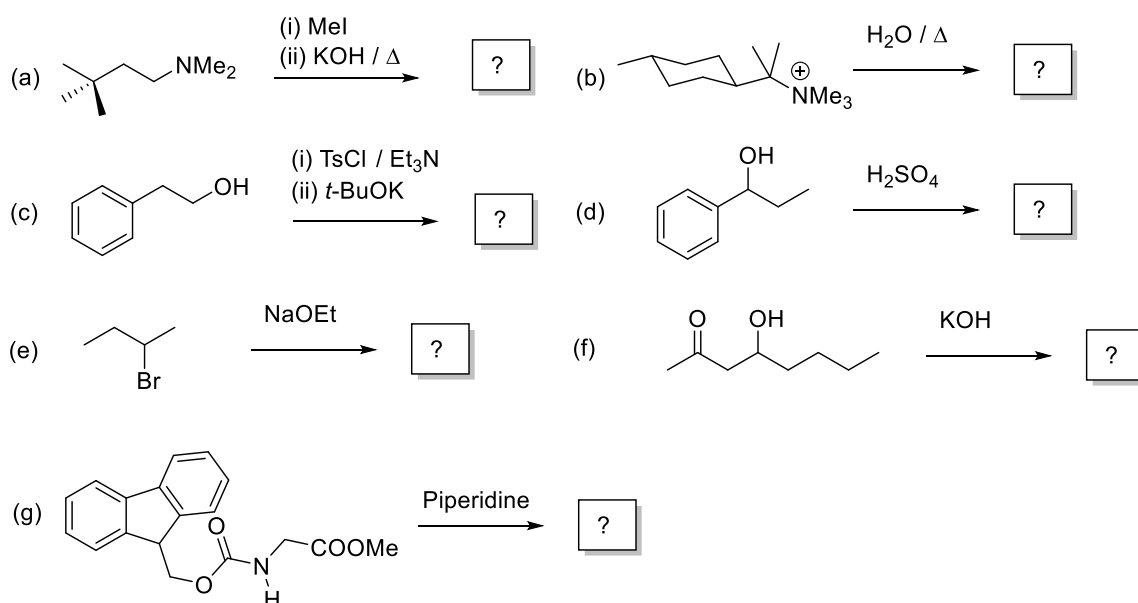
CHM102 Basic Organic Chemistry

Problem Set 4: Reactive intermediates / Reactivity / Selectivity & Specificity

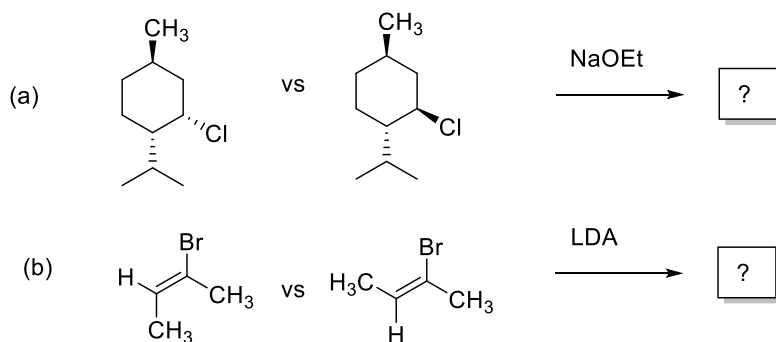
Q1. Identify the product and the reactive intermediate(s) involved in the reactions. Also, indicate arrow-push mechanism for the product formation.



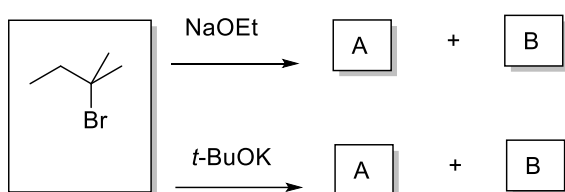
Q2. Indicate which of the following reactions can dominantly happen through one among E_1 , E_2 and E_{1cb} mechanisms. Identify the factor responsible for the choice of the mechanism in each case. Also, indicate arrow-push mechanism for the product formation.



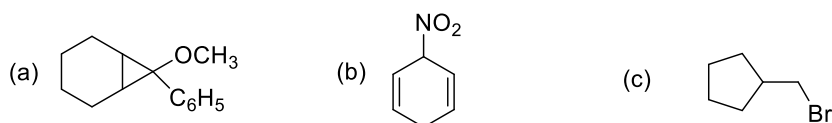
Q3. Which among the following pairs will undergo elimination reaction faster? Predict the product, if any and indicate the reason.



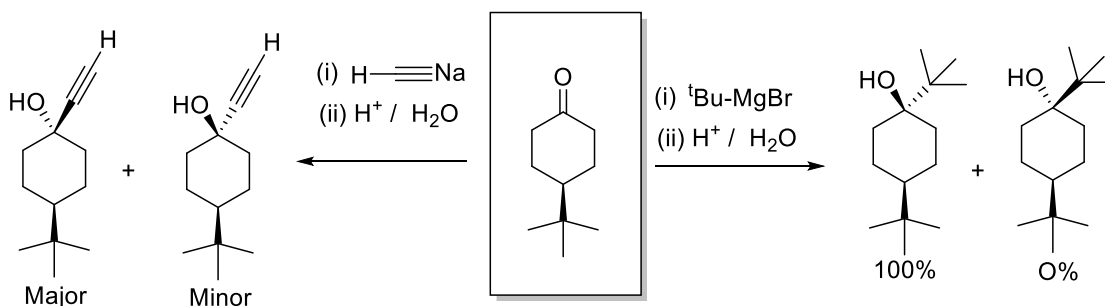
Q4. Identify the products "A and B" and predict the product distributions (by indicating the major and minor products) for the following reactions. Identify the reason.



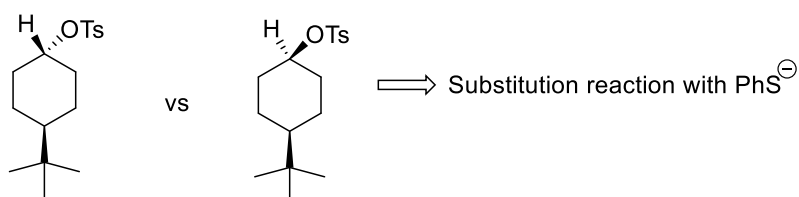
Q5. Provide a synthetic route for accessing the following compounds. Consider any starting material, reactive intermediate(s) and indicate appropriate reaction conditions.



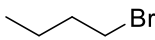
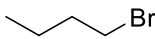
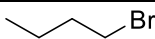
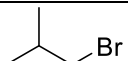
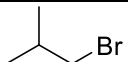
Q6. Explain the reason in nucleophilic addition to the following carbonyl compound show different stereoisomeric products with different nucleophiles? Also, classify those reactions between stereoselective/stereospecific/regioselective/regiospecific reactions.



Q7. Which among the following substrates reacts faster with the given nucleophile? Explain the mechanism with reason.

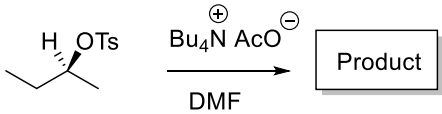
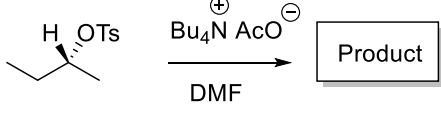
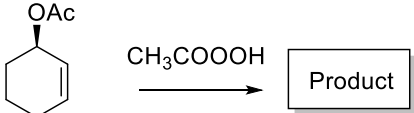
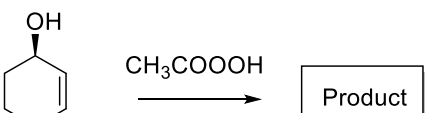
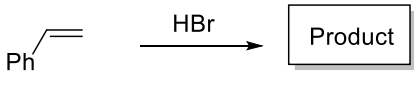
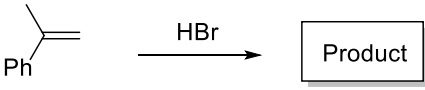
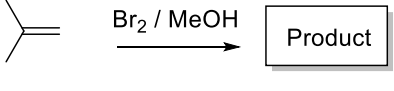
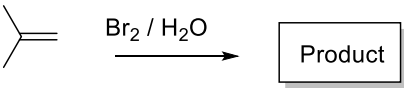


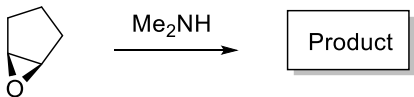
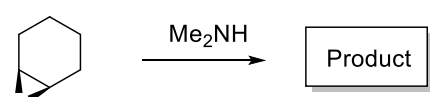
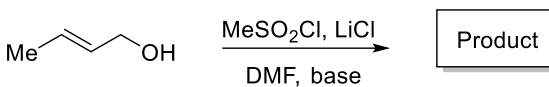
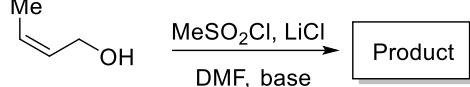
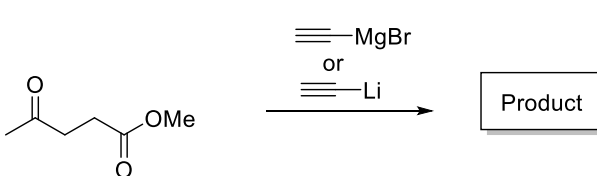
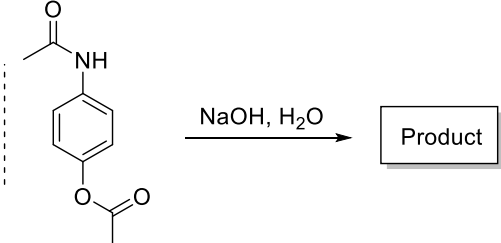
Q8. Predict the major product for the following reaction, classify the reaction based on the mechanism and indicate the reason.

Reactant	Reaction condition	Product	Type of mechanism	Reason
	NaCN			
	NaOCH ₃			
	KO ^t Bu ₃			
	NaI			
	NaOCH ₃			
	EtOH			
	CH ₃ SNa			
	NaOCH ₃			
	EtOH			
	NaOCH ₃			

Q9. How to prepare and syn and anti 1,2-diols can be synthesized from a common olefin? Explain the mechanism.

Q10. Predict the product and classify the following reaction(s) as stereoselective / stereospecific / regioselective / regiospecific / chemoselectivity with reasons (including arrow-push mechanism).

(a)		
(b)		
(c)		
(d)		

(e)		
(f)		
(g)		
(h)	